



Chemistry Department Organic Chemistry 217

كيمياء عضوية
ك 217
د. حسين المغيض

Chapter 1

Electronic structure and covalent bonding

Polar Covalent Bond



Dr. Hussein Al-Mughaid

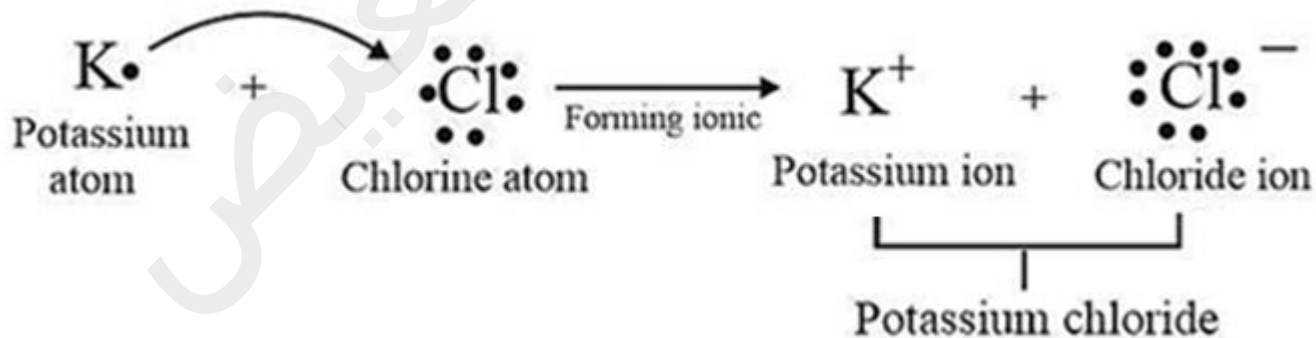
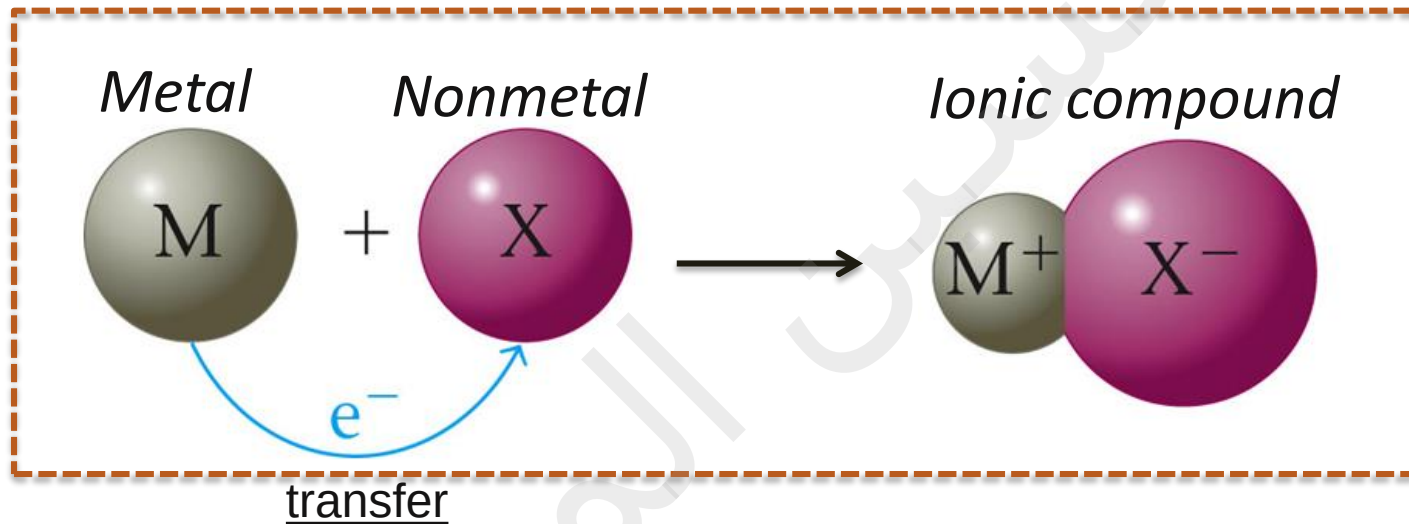
Lewis Symbols and the Valence Concept

- Valence electrons = **Group number**.
- Lewis dot symbols** represent the valence electrons of an atom as dots arranged around the atomic symbol.

1A 1	2A 2	Electron dot-formula										3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
•H												•B•	•C•	•N•	•O•	•F•	He•
•Li	•Be•											•Al•	•Si•	•P•	•S•	•Cl•	•Ar•
•Na	•Mg•	3B 3	4B 4	5B 5	6B 6	7B 7	8B 8	9	10	1B 11	2B 12	•Ga•	•Ge•	•As•	•Se•	•Br•	•Kr•
•K	•Ca•											•In•	•Sn•	•Sb•	•Te•	•I•	•Xe•
•Rb	•Sr•											•Tl•	•Pb•	•Bi•	•Po•	•At•	•Rn•
•Cs	•Ba•																
•Fr	•Ra•																

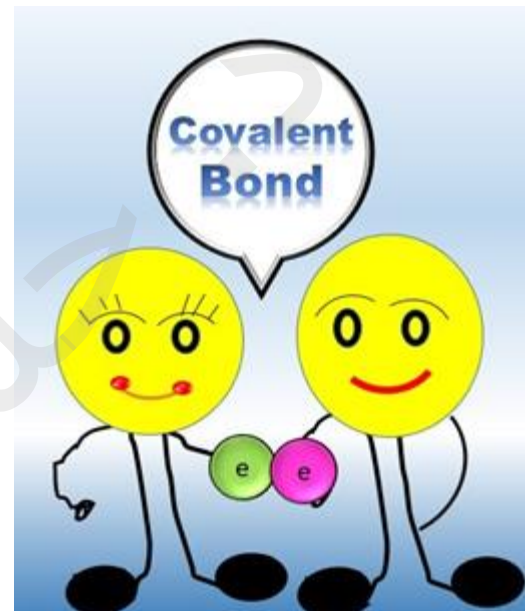
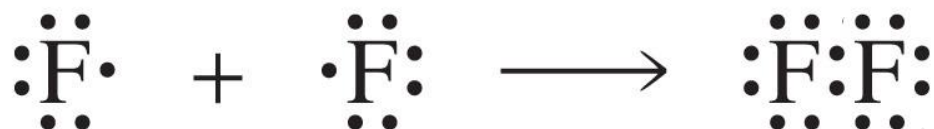
Ionic Bond

- It is basically the transfer of one electron from one atom to another and based on electrostatic forces (attractions) between two oppositely-charged ions

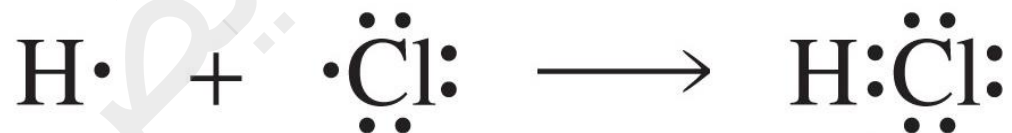


Covalent Bonds are Formed by Sharing Electrons

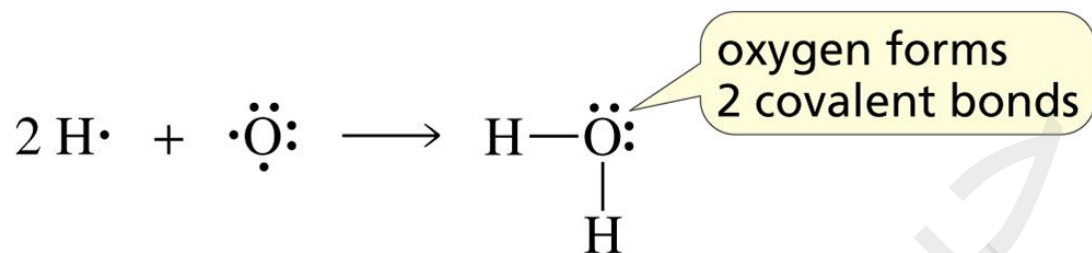
Nonpolar: **bonded atoms** are the same



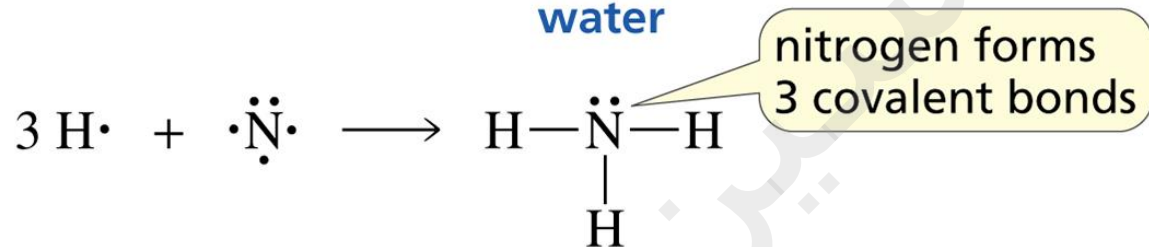
Polar: **bonded atoms** are different



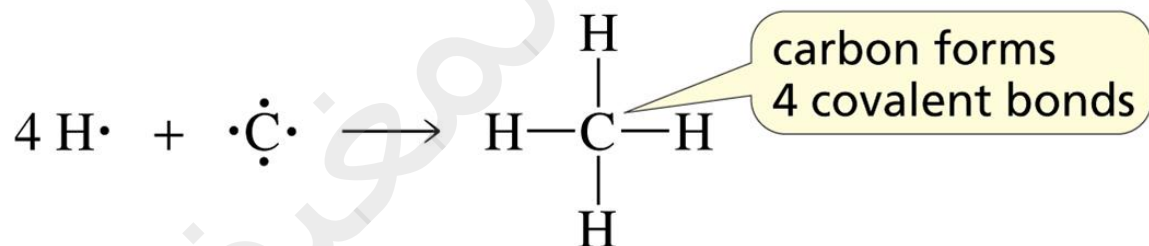
How Many Bonds Does an Atom Form?



water



ammonia



methane

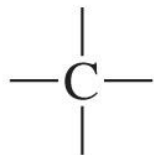
notice that each O, N, C is surrounded by 8 electrons, and each H is surrounded by 2 electrons

The Number of Bonds and Lone Pairs



1 bond

of bonds +
of lone pairs



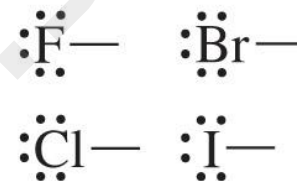
4 bonds
 $\frac{0 \text{ lone pairs}}{4}$



3 bonds
 $\frac{1 \text{ lone pair}}{4}$

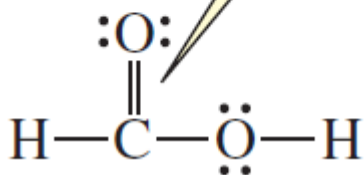


2 bonds
 $\frac{2 \text{ lone pairs}}{4}$



1 bond
 $\frac{3 \text{ lone pairs}}{4}$

2 covalent bonds holding 2 atoms together is called a double bond



3 covalent bonds holding 2 atoms together is called a triple bond

Bond Polarity Depends on Electronegativity Differences(ΔX)

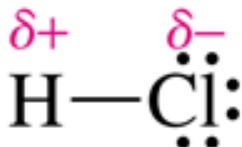
H 2.1								
Li 1.0	Be 1.5		B 2.0	C 2.5	N 3.0	O 3.5	F 4.0	
Na 0.9	Mg 1.2		Al 1.5	Si 1.8	P 2.1	S 2.5	Cl 3.0	
K 0.8	Ca 1.0						Br 2.8	
							I 2.5	

 **N-H** bond is less polar ($\Delta X = 0.9$), than **O-H** bond ($\Delta X = 1.4$).

 **NaCl** ($\Delta X = 2.1$), is not as ionic as **KF** ($\Delta X = 3.2$).

How to express the polarity of a polar covalent bond

1. Delta notation:



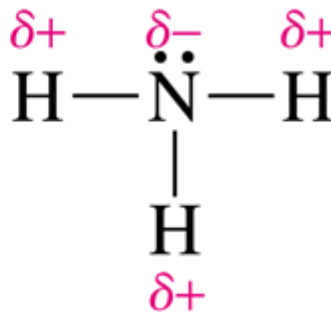
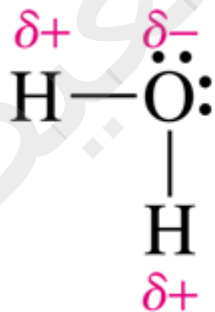
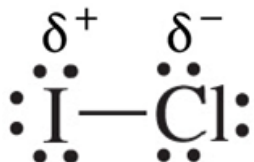
a slight positive charge on one end and
a slight negative charge on the other

2. Cross-based arrow:



the negative end
of the bond

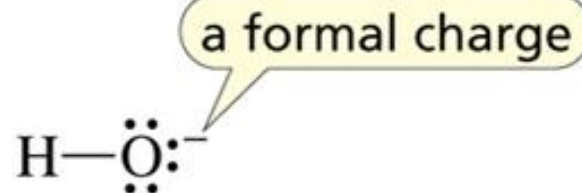
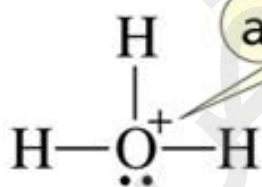
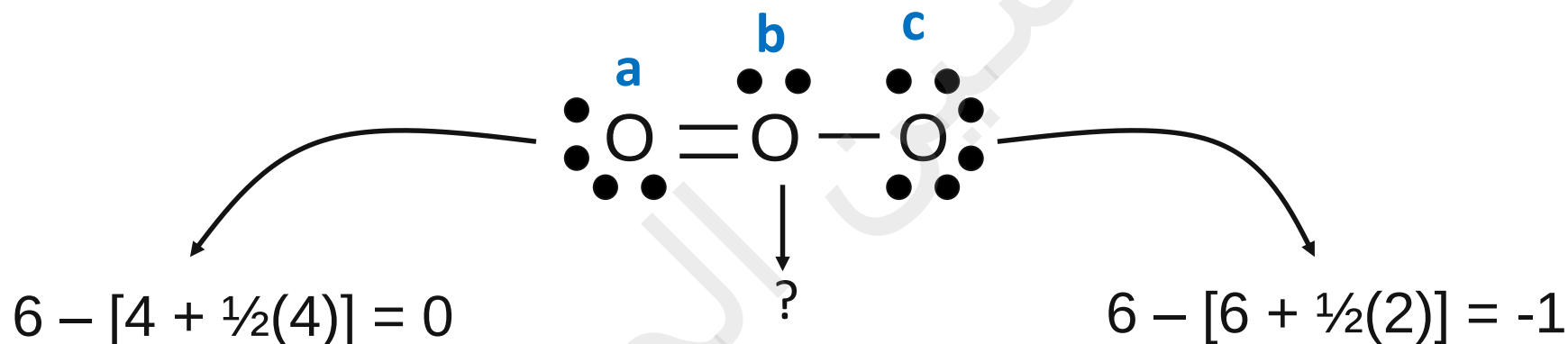
More examples



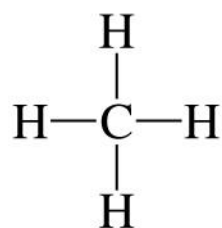
Formal Charges

Formal charge = number of valence electrons - (number of **lone-pair** electrons + $\frac{1}{2}$ number of bonding electrons)

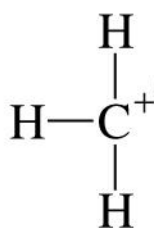
Calculate the formal charge on each atom in O_3 .



Carbon Forms Four Bonds

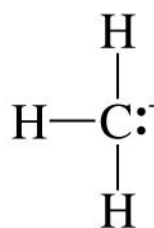


methane



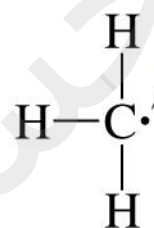
methyl cation

a carbocation



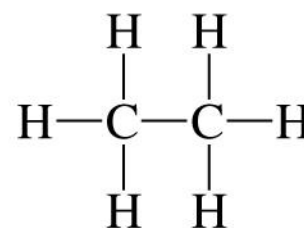
methyl anion

a carbanion



methyl radical

a radical

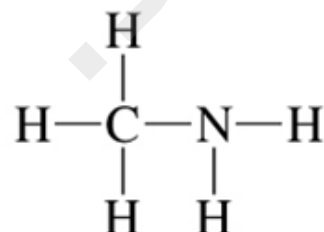
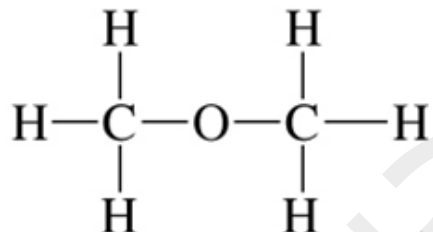
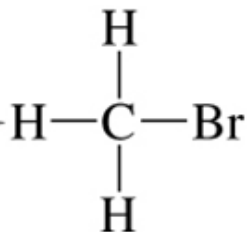


ethane

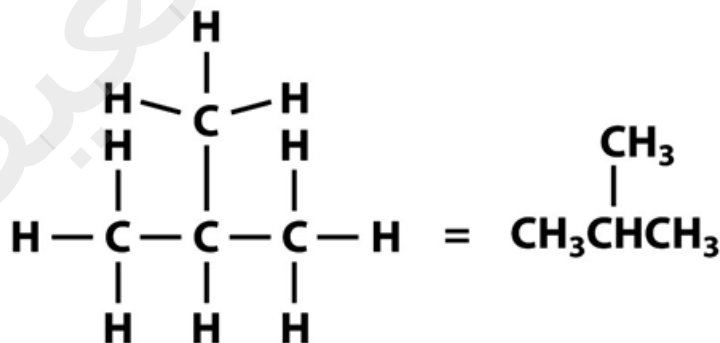
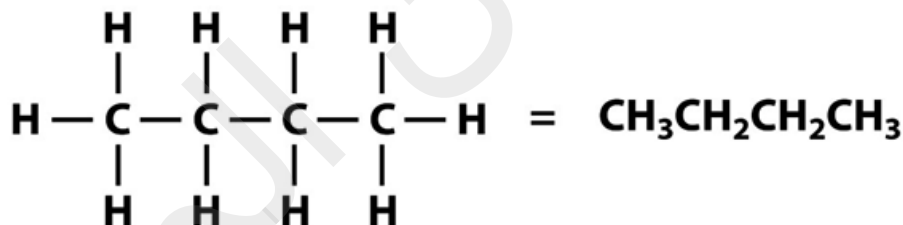
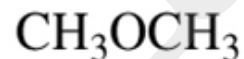
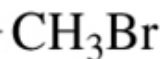
If carbon does **not** form **four** bonds,
it has a **charge** (or it is a **radical**).

Kekulé Structures and Condensed Structures

Kekulé structures



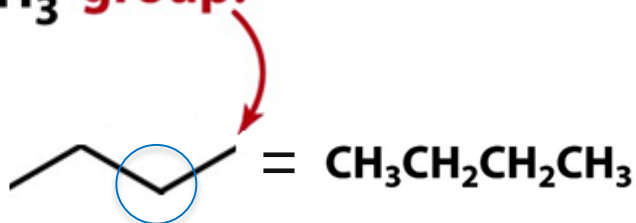
condensed structures



Skeletal Structures

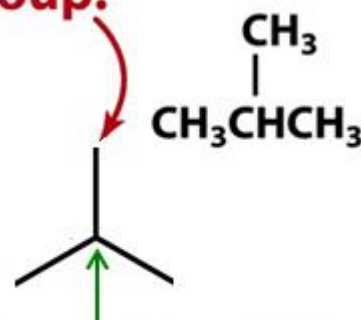
1. Show the carbon-carbon bonds as lines
2. Do not show the **carbons** or the **hydrogens** that are bonded to the carbons.

This is a CH₃ group.



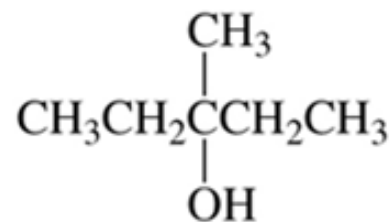
**This intersection
is a CH₂ group.**

This is a CH₃ group.

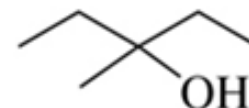
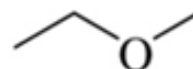


**This is a CH group.
three-way intersection**

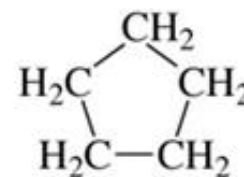
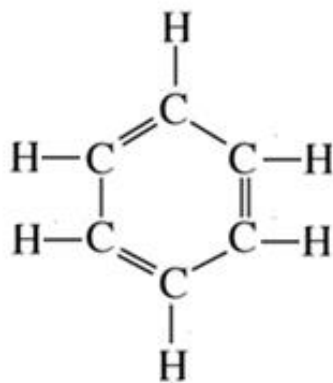
condensed
structures



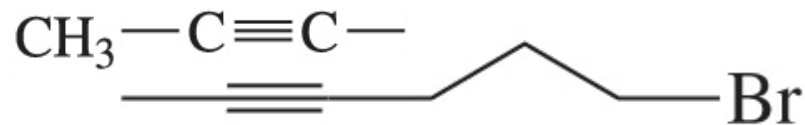
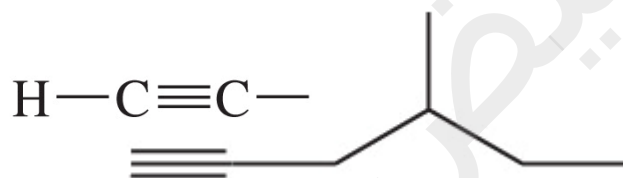
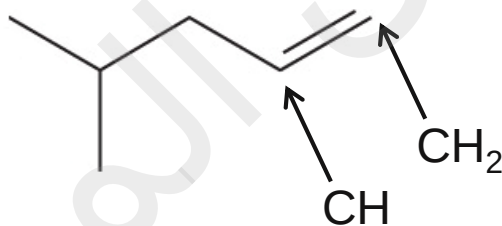
skeletal
structures



condensed structures

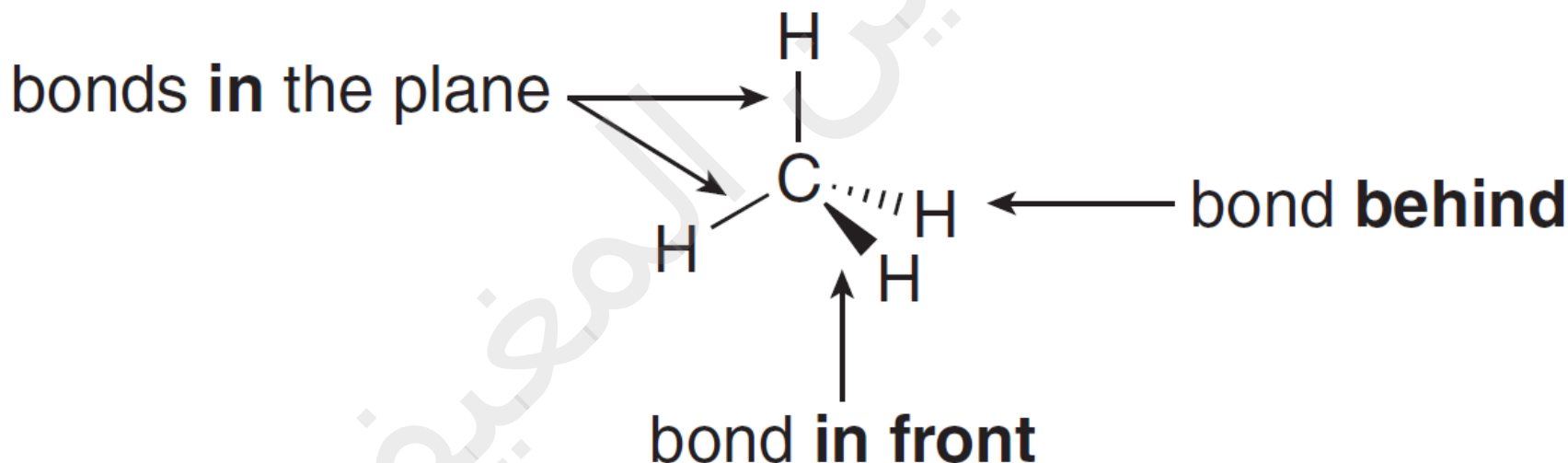


skeletal structures



three-dimensional geometry of a tetrahedron on a two-dimensional piece of paper

- A *solid line* is used for a bond *in* the plane.
- A *wedge* is used for a bond *in front* of the plane.
- A *dashed wedge* is used for a bond *behind* the plane.

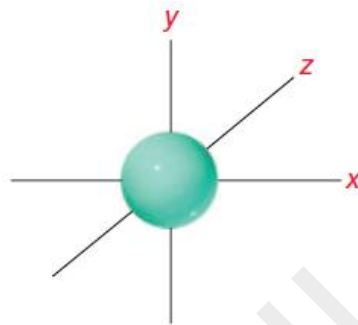


Atomic orbitals

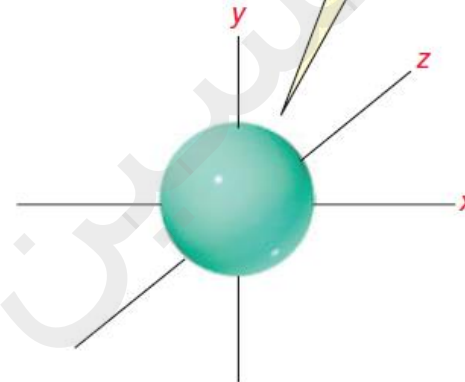
An atomic orbital is the three dimensional region around the nucleus where an Electron is most likely to be found.

1. The s Orbital

resemble spheres



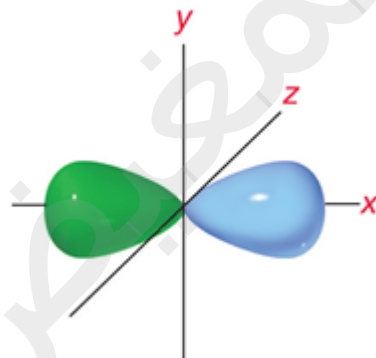
1s orbital



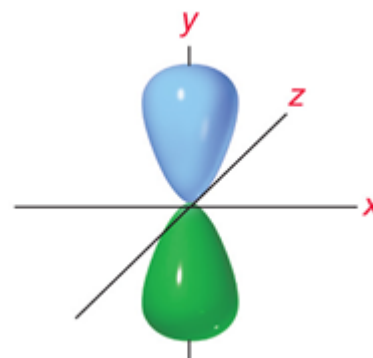
2s orbital

2. The P Orbital

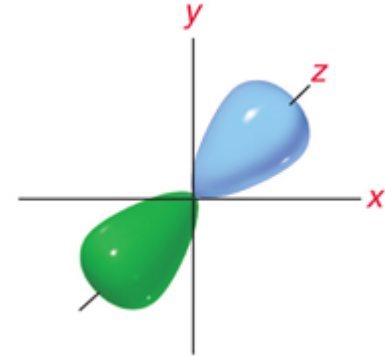
defined by two lobes
(teardrop shaped)



$2p_x$ orbital

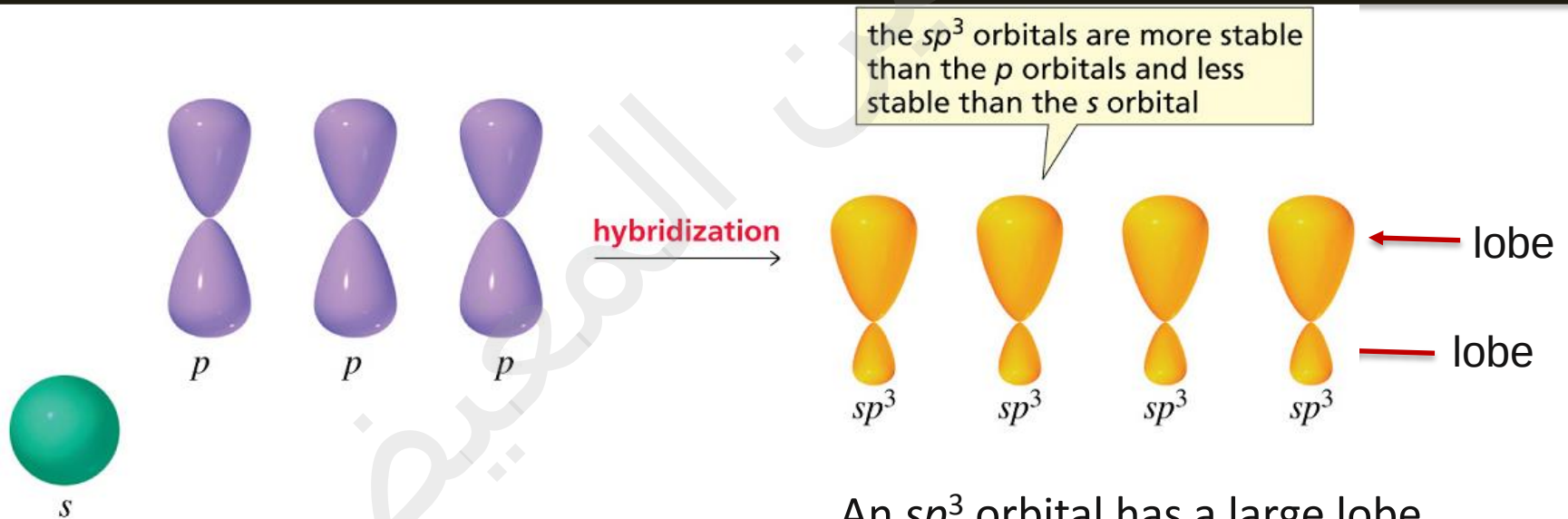
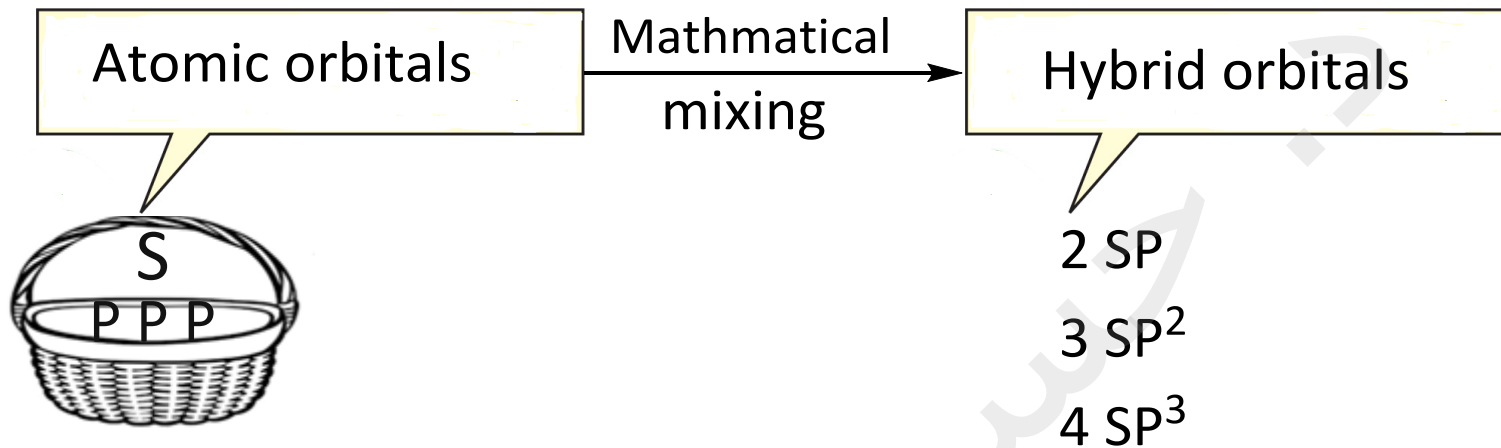


$2p_y$ orbital



$2p_z$ orbital

Hybridization



An sp^3 orbital has a large lobe and a small lobe.

How can I figure out the hybridization of a given atom?



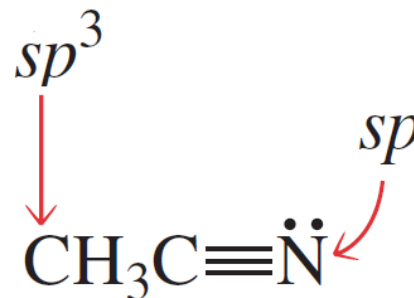
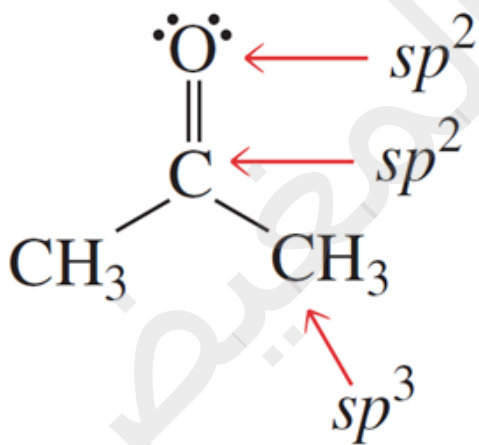
Determine the Steric number

Steric number = number of sigma bonds + number of lone pairs

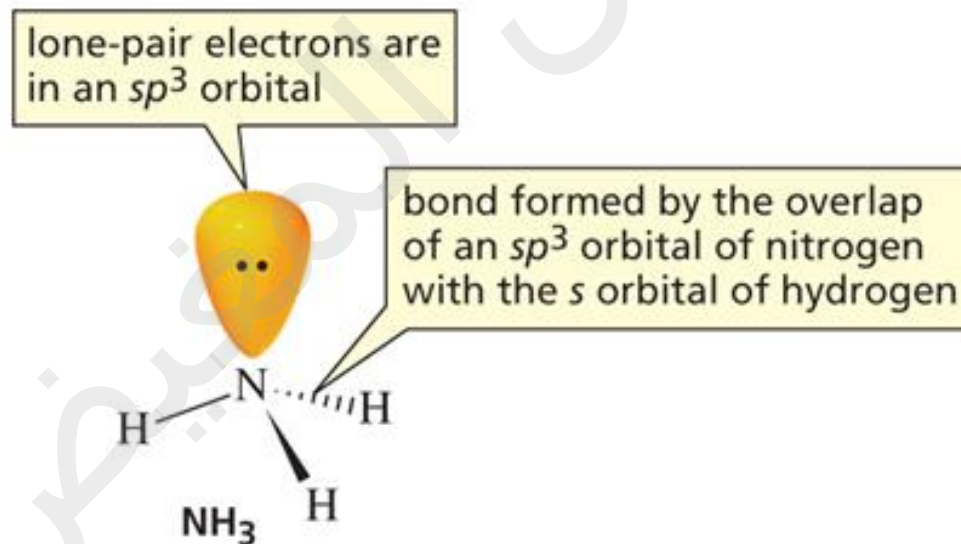
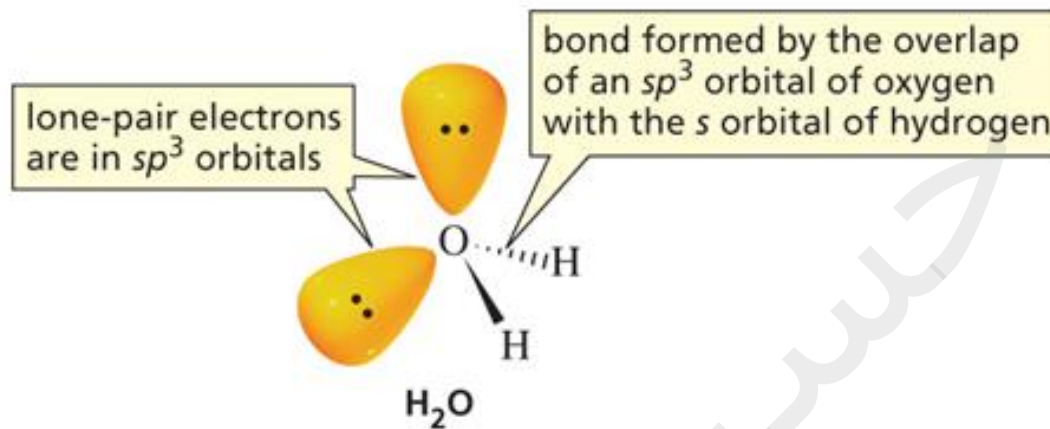
- If the Steric number is **4**, then it is **sp^3**
- If the Steric number is **3**, then it is **sp^2**
- If the Steric number is **2**, then it is **sp**

Note

double bond count as a single bond, as do triple bond



In what orbitals are the lone pairs?



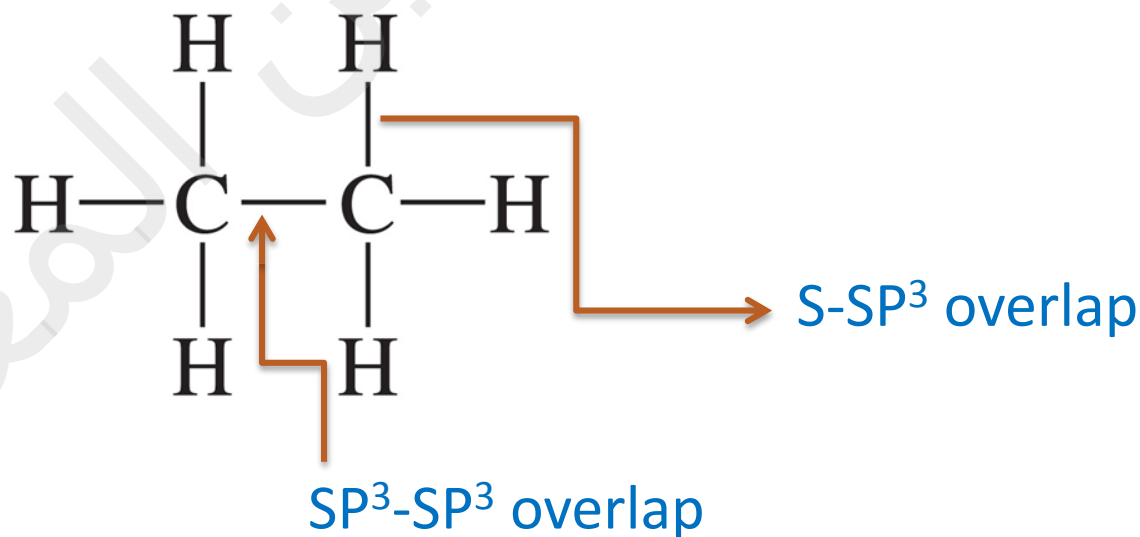
Summary of Hybridization and Geometry

Hybrid Orbitals	Hybridization	Geometry	Approximate Bond Angles
2	$s + p = sp$	linear	180°
3	$s + p + p = sp^2$	trigonal	120°
4	$s + p + p + p = sp^3$	tetrahedral	109.5°



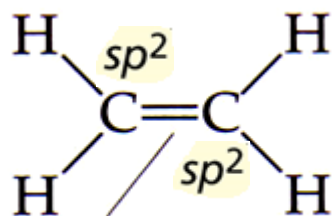
Carbon used hybrid orbitals (sp , sp^2 , sp^3) to form sigma (σ) bonds

**Look How the C-C
and C-H bonds are
formed**

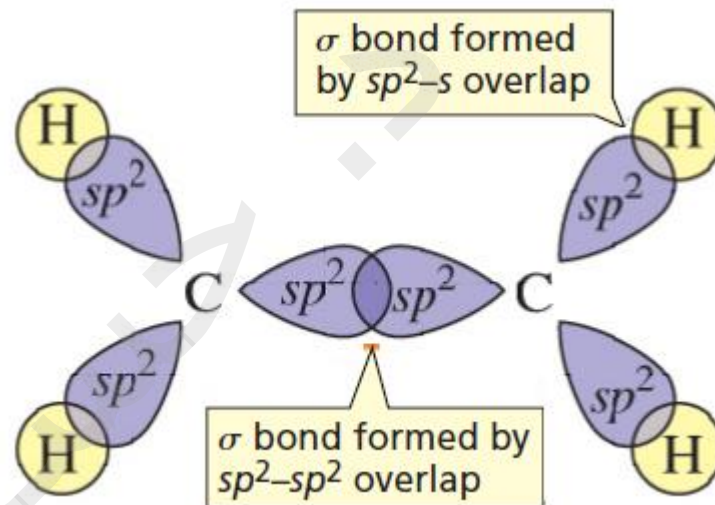


Multiple Bonds

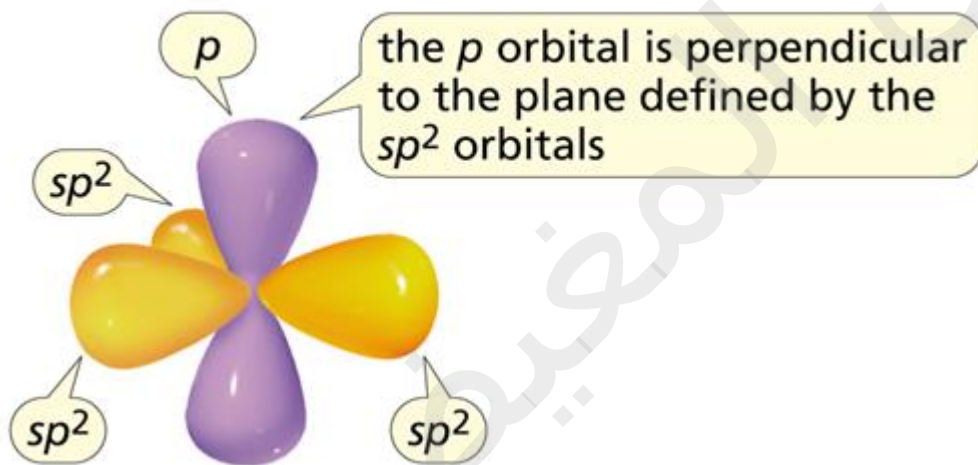
1. Double Bond:



a double bond consists of one σ bond and one π bond



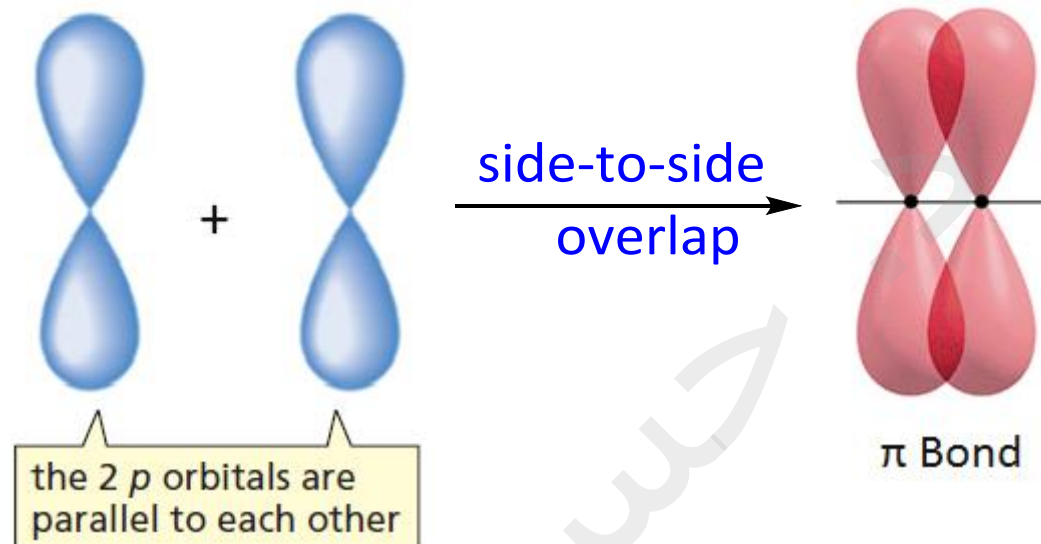
σ bond framework



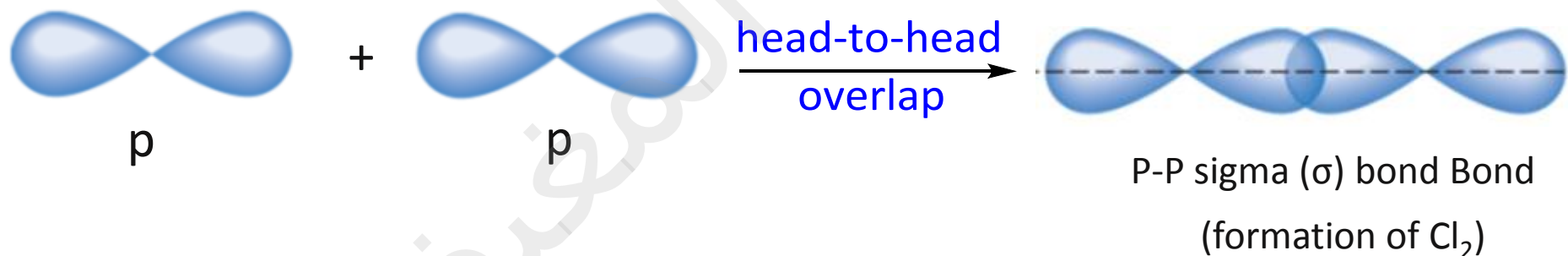
An sp^2 Carbon Has Three sp^2 Orbitals
and One p Orbital



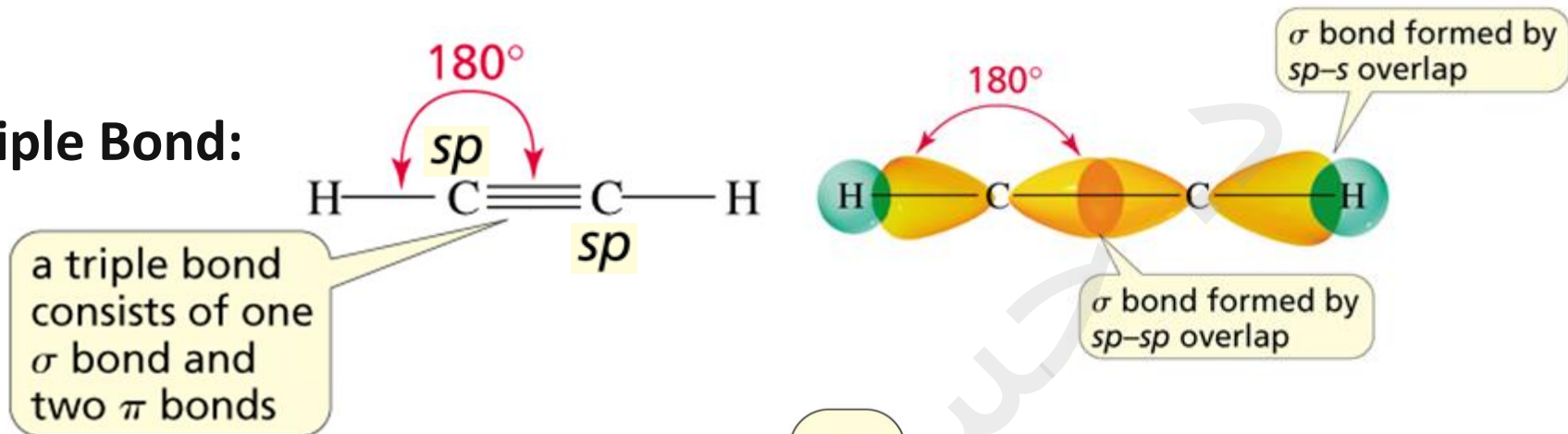
How the $\pi(\pi)$ bond form?



Note



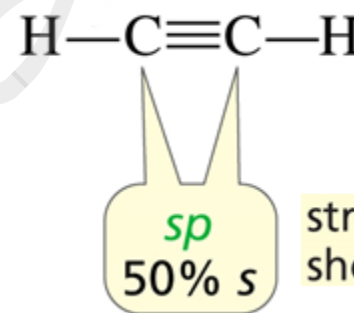
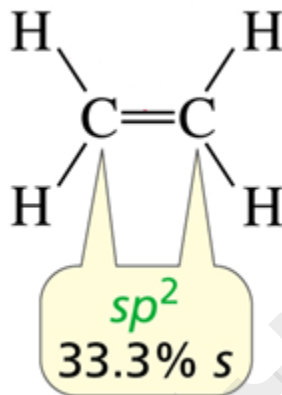
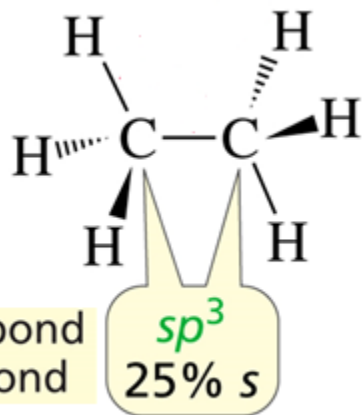
2. Triple Bond:



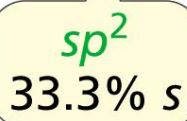
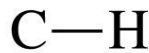
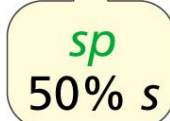
The Two sp Orbitals Point in Opposite Directions; The Two p Orbitals are Perpendicular

Bond Length and Bond Strength: Hybridization effect

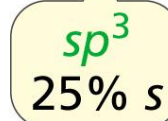
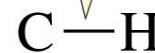
The **more s character**, the **stronger** and **shorter** is the bond.



strongest bond
shortest bond



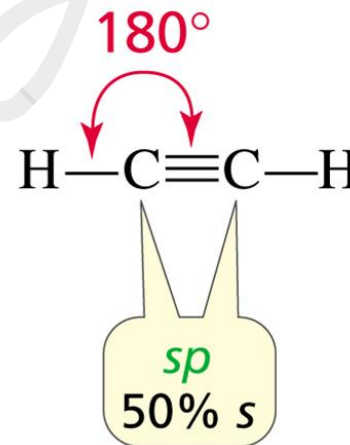
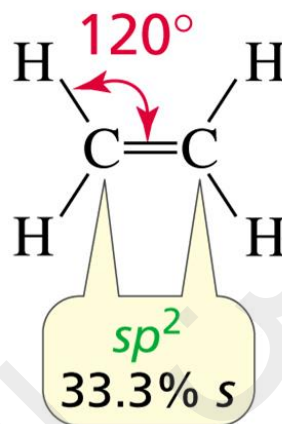
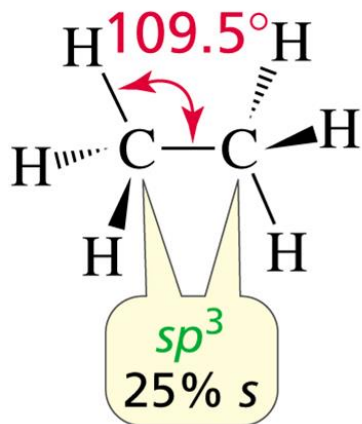
weakest bond
longest bond



bond strength increases as bond length decreases

Hybridization Affects the Bond Angle

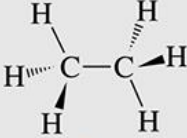
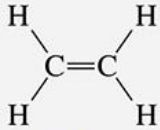
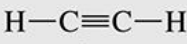
The more *s* character, the greater the bond angle.



bond angle increases as *s* character in the orbital increases

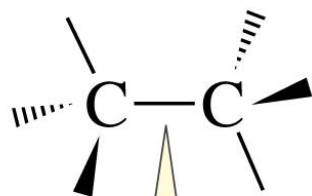
Summary: Hybridization, Bond Angle, Bond Length, Bond Strength

Table 1.7 Comparison of the Bond Angles and the Lengths and Strengths of the Carbon–Carbon and Carbon–Hydrogen Bonds in Ethane, Ethene, and Ethyne

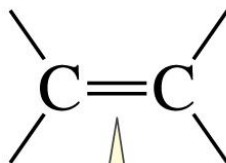
Molecule	Hybridization of carbon	Bond angles	Length of C—C bond (Å)	Strength of C—C bond (kcal/mol)	Length of C—H bond (Å)	Strength of C—H bond (kcal/mol)
 ethane	sp^3	109.5°	1.54	90.2	1.10	101.1
 ethene	sp^2	120°	1.33	174.5	1.08	110.7
 ethyne	sp	180°	1.20	230.4	1.06	133.3

Summary

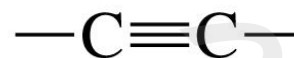
- The **shorter** the bond, the **stronger** it is.
- The **more s character**, the **shorter** and **stronger** the bond.
- The **more s character**, the **larger** the bond angle.



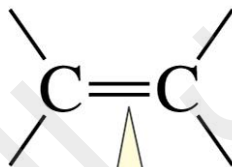
one σ bond



one σ bond
one π bond



one σ bond
two π bonds

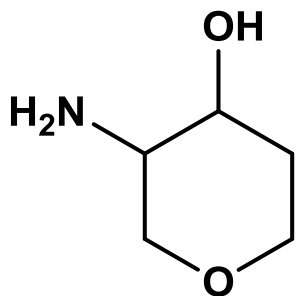


strength of the double bond = 174 kcal/mol
strength of the sp^2-sp^2 σ bond = -112 kcal/mol
strength of the π bond = 62 kcal/mol

a π bond is weaker than a σ bond

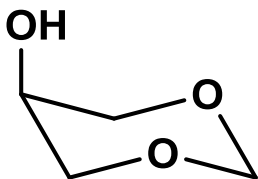
PRACTICE the skill

Q1) How many polar covalent bonds does the compound shown below have?



<C> 4 <C> 5 <C> 6 <C+> 7

Q2) How many polar covalent bonds does the compound shown below have?

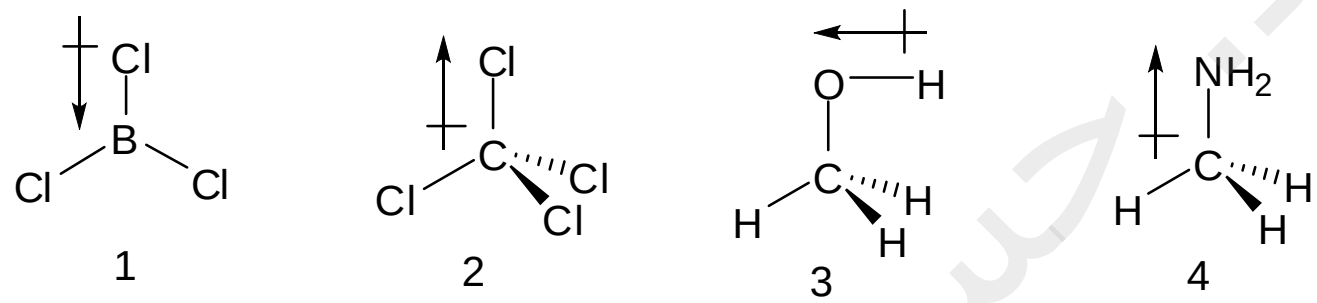


<C> 3 <C> 5 <C+> 6 <C> 7

Q3) The molecule H_3C-X is most polar when X is:

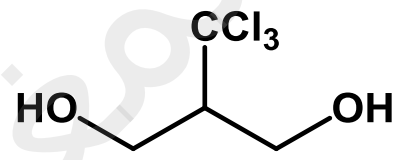
<C> Cl <C+> F <C> CH_3 <C> OH

Q4) Which of the following drawings of the direction of the indicated bond polarity is incorrect?



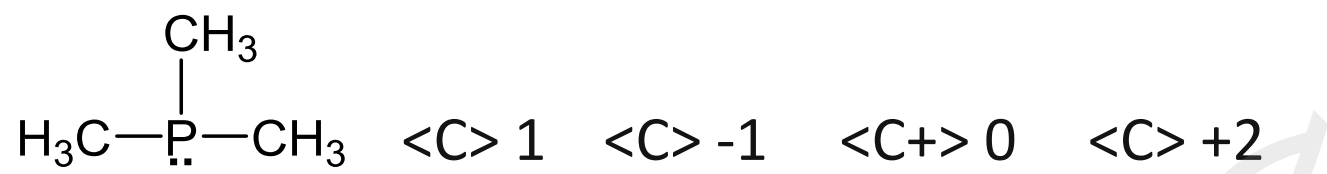
<C+> 1 <C> 2 <C> 3 <C> 4

Q5) When you apply delta notation($\delta+$, $\delta-$) to all bonds in the following molecule, how many atoms will carry partially plus($\delta+$)?

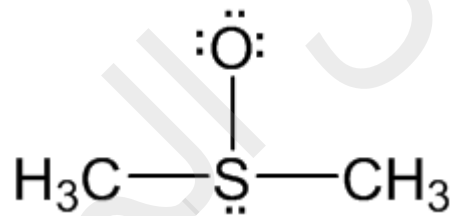


<C+> 5 <C> 2 <C> 3 <C> 4

Q6) The formal charge on phosphorus (P, group 5A) in the following molecule is:



Q7) For the following compound, indicate the formal charges for Sulfur (S, group 6A) and oxygen (O, group 6A)?



- I) S = zero, O = -1

II) S = -1, O = zero

III) S = +1, O = -1

IV) S = -1, O = +1

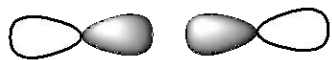
<C> I

<C> II

<C+> III

<C> IV

Q8) As drawn, which pair of orbitals when brought together, will not form sigma bond?



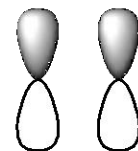
1



2



3



4

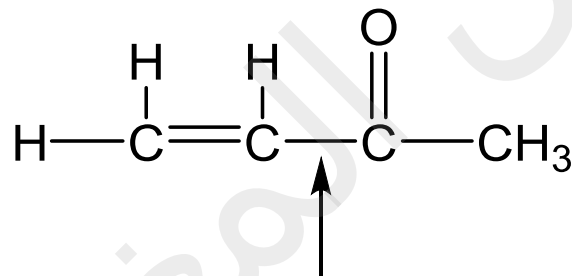
<C> 1

<C> 2

<C+> 4

<C> 3

Q9) In the compound shown below, how the indicated bond formed?



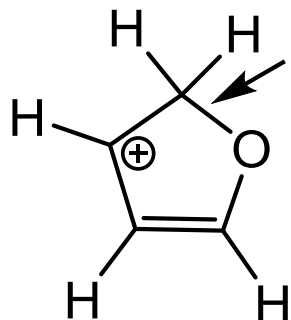
<C+> sp^2 - sp^2 overlap

<C> sp^3 - sp^3 overlap

<C>) sp - sp^3 overlap

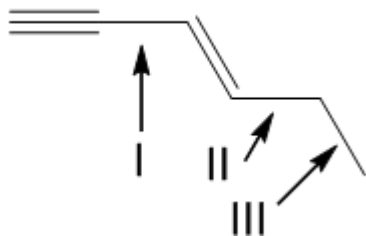
<C> sp - sp^2 overlap

Q10) In the compound shown below, how the indicated bond formed?



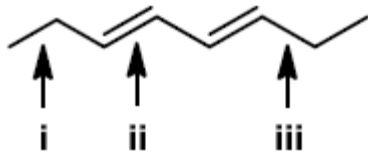
- <C> sp^3-sp^2 overlap.
- <C> sp^2-sp^2 overlap.
- <C> sp^3-sp overlap.
- <C+> sp^3-sp^3 overlap

Q11) Which of the following indicated C—C bonds is(are) the shortest?



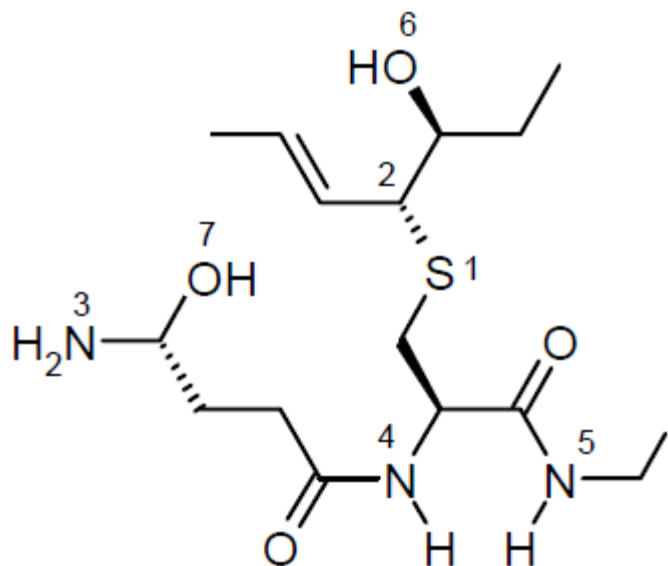
- <C+> I
- <C> II
- <C> III
- <C+> both I and II

Q12) The relative strengths of the indicated CC bonds is



- A) $i > ii > iii$ B) $ii > iii > i$
- C) $i > iii > ii$ D) $iii > i > ii$
- E) $ii > i > iii$

*The questions 13-16 are about the compound shown below



Q13) What is the C-S-C bond angle for **S1**?

A) 90° B) 180° C) 109.5° D) 120°

Q14) For **N3**, what is the H-N-H bond angle?

A) 90° B) 180° C) 109.5° D) 120°

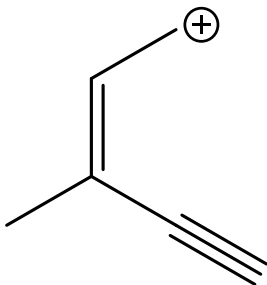
Q15) How many sp^2 hybridized carbon atoms are there in the above compound?

A) 2 B) 3 C) 4 D) 5

Q16) What are the hybridizations of **S1** / **O6** ?

A) sp^3 / sp^3 B) sp^3 / sp^2 C) sp^2 / sp^2 D) sp / sp E) sp / sp^3

Q17) *in the following species, how many p-orbitals are there?*

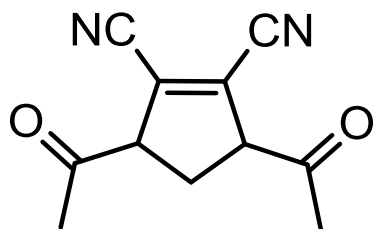


- A) 6 B) 3 C) 7 D) 8

Q18) *The element Br can form the same number of bonds as:*

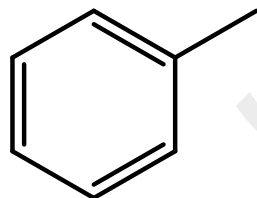
- A) H B) N C) O D) S

Q19) *How many π bonds does the compound shown below have?*



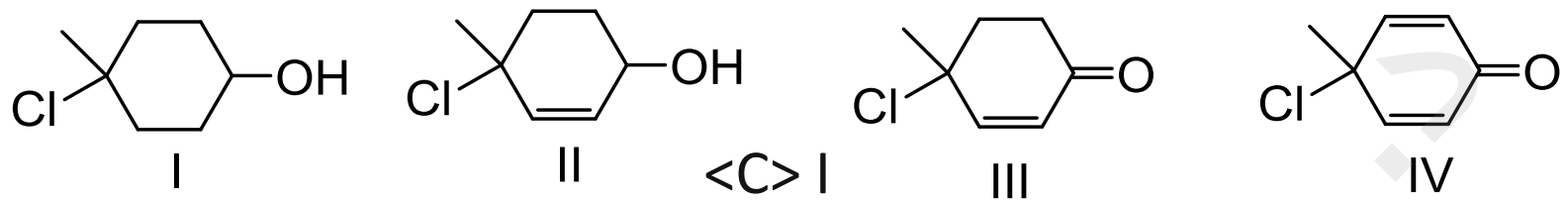
- A) 3 B) 5 C) 1 D) 7

Q20) *How many σ bonds does the compound shown below have?*



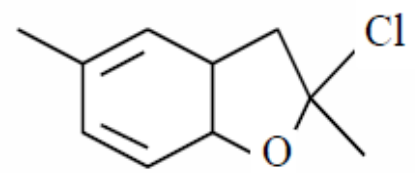
- A) 3 B) 7 C) 12 D) 15

Q21) Which of the following molecules contains 7 hydrogen atoms?



- <C> I
- <C> II
- <C> III
- <C+> IV

Q22) How many hydrogen atoms are there in the compound shown?



- <C+> 13
- <C> 14
- <C> 15
- <C> 16

Q23) what's orbital overlap to create the C-H bond in $^+CH_3$?

- <C> SP^3 - SP^2 overlap.
- <C+> S- SP^2 overlap.
- <C> S-SP overlap.
- <C> S- SP^3 overlap

